```
!pip install pyspark py4j
# Import pyspark
from pyspark.sql import SparkSession
```

```
# CreateSpark Session
spark = SparkSession.builder.appName("Session01").getOrCreate()
```

```
# Read CSV file into a DataFrame
filePath = "/content/sample_data/Data_Train.csv"
df = spark.read.csv(filePath, header=True, inferSchema=True)
```

```
# Show the first few rows of the DataFrame
df.show()
```

```
#Print only 1st 4 Rows
df.head(4)
```

```
# Know the DataType
type(df)
pyspark.sql.dataframe.DataFrame
```

```
# Know the DataType
df.dtypes
[('Airline', 'string'),
('Date_of_Journey', 'string'),
('Dep_Time', 'timestamp')]
```

```
# printSchema
df.printSchema()
|-- Airline: string (nullable = true)
|-- Date_of_Journey : string (nullable = true)
|-- Dep_Time : timestamp (nullable = true)
```

```
#print columns
df.columns
```

```
# print column/columns
df.select("Airline").show()
df.select(["Airline","Date_of_Journey"]).show()
```

```
# Describe dataframe
df.describe().show()
```

```
# Add New Columns
df1 = df.withColumn("Arrival_MinuteNew",df['Arrival_Minute']+10)
```

```
# Drop A Column/Columns
df.drop('Arrival_MinuteNew').show()
df.drop('Arrival_Time','Dept_Minute').show()
```

```
# Rename the column
df.withColumnRenamed('Airline',"Flight").show()
```

```
# Drop Na
df.na.drop(how='any',thresh=1).show()
# drop na from specific column
df.na.drop(subset=['Dept_Hour']).show()
```

```
# fill na with specific value
df.na.fill(10).show()
```

```
# Fill na with some value
from pyspark.sql.functions import mean
meanVlaue = df.select(mean(df['Dept_Hour'])).collect()[0][0]
df.na.fill(meanVlaue).show()
```

```
# Fliter operation
df.filter(df["Dept_Hour"]<=9).show()
```

```python
# Filter and print only 'Dep_Time','Arrival_Time'
df.filter(df["Dept_Hour"]<=9).select('Dep_Time','Arrival_Time').sh
ow()

# filter with two conditions
df.filter((df['Dept_Hour']<=12) | (df['Arrival_Hour']<=5)).show()
```

```python
## Group By and Aggregate Function
df.groupBy('Airline').count().show()
df.groupBy('Airline').sum().show()
df.groupBy('Airline').mean().show()
```

```python
# Create a list of dictionaries containing sample data
data = [{"Name": "John", "Age": 25},
        {"Name": "Alice", "Age": 30},
        {"Name": "Bob", "Age": 35}]
# Create DataFrame using the list of dictionaries
df = spark.createDataFrame(data)
```

```python
from pyspark.sql import SparkSession
# Create a SparkSession
spark =
SparkSession.builder.appName("JoinExample").getOrCreate()
```

```python
# Sample data for DataFrame1
data1 = [("Alice", 1),("Charlie", 2),("Bob", 3)]
columns1 = ["name1", "value1"]
df1 = spark.createDataFrame(data=data1, schema=columns1)
df1.show()
```

```python
# Sample data for DataFrame2
data2 = [("Alice", 10), ("Bob", 20), ("David", 30)]
columns2 = ["name2", "value2"]
df2 = spark.createDataFrame(data=data2, schema=columns2)
df2.show()
```

```python
# Joining DataFrames
comCol = df1['name1']==df2['name2']
df1.join(df2,on=comCol,how='inner').show()
```

```python
# Left Anti Join Example
df1.join(df2,on=comCol,how='left_anti').show()
```

```python
# Left Semi Join Example
df1.join(df2, on=commonColumns, how="left_semi").show()
```

Create new Branch

```
git branch <branch-name>
```

Create and Switch to New Branch

```
git checkout -b <branch-name>
git switch -c <branch-name>
```